Sustainability Economics: Complete Lecture Notes

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Course Overview & Syllabus Summary

What is the connection between economics and sustainability? When people talk about economics, often they are talking about something to do with money. How people acquire it and what they do with it. When people talk about sustainability, often they are talking about the environment, how it is changing, and whether or not those changes are good. Where do these two things come together? That intersection often seems to be about the way that people are acquiring and using money is changing the environment in ways that are not good.

Part I

Foundations & Basic Principles

Chapter 1

Introduction & Key Concepts

1.1 Overview & Objectives

The term *sustainability* implies continuity and balance. If something is sustainable, it can continue doing what it is doing. Unless it is doing nothing, what it is doing is interacting with other entities. This implies that, except for those things that are doing nothing, we should think about sustainability with a systems perspective.

A system is a collection of entities (or elements) and the connections between them. To understand a system, we need to understand the entities and their connections. Our goal in this course is to develop an understanding of the relationship between the economy and the environment by looking at the elements that make up the economy and their relationship with elements that make up the environment. To do so, we will proceed by building a model of a *complex adaptive social ecological system*.

The system we inhabit, and that we will try to understand, is complex. This means that the system as a whole exhibits properties that cannot be explained by looking at any element itself. It is adaptive in that it changes in response to internal and external pressures.

ey reading for this section: Chapter 1 of the required textbook.

Slide 1: Systems

- A system is a collection of elements and their connections.
- A social ecological system contains elements in both the social (human) and ecological (nature) domains, and has connections crossing between the domains.
- A system is complex when there are 'emergent' properties that cannot be predicted looking at elements in isolation.
- A system is adaptive if it continuously evolves or changes in response to internal and external forces.

Slide 2: Systems



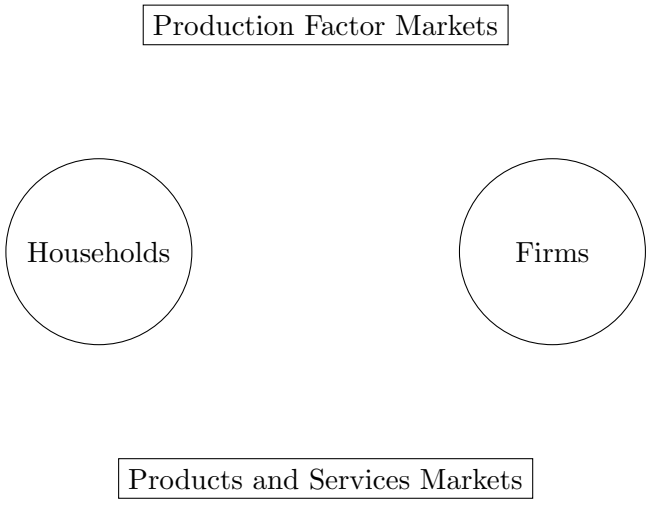
1.1.1 Systems and Subsystems

The global social ecological system is far too complex for anyone to comprehend. At present, it is probably too complex for AI to comprehend as well. What both humans and AI do with this is focus on a part of the system that we are interested in. We choose a boundary and a scale. Things beyond the boundary are grouped together into subsystems, the internal details of which are not considered. This would be at the scale that is coarser than the part of the system we are interested in. For those things at a finer scale, we also relegate those things into subsystems, with the entire subsystem being taken as an element of the system we are studying.

The concerns about sustainability can be seen as concerns about the details that are being left out.

Slide 3: Production Possibilities

- What is missing?



Chapter 2

Systems

2.1 Systems

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Slide 5: Systems



When we think about systems, we need to consider scale. If we consider the scale of ...

Slide 6: Scales

- What constitutes elements of a system depends on the physical scale at which we are contemplating the system.
  - At the atomic scale, atoms are the elements.
  - At the cosmic scale, galaxies are the elements.
  - What are the elements at the human scale?
- What the connections between elements are depends on the physical scale and the temporal scale chosen.
  - How long does it take for the connection to change something for the elements that we are interested in?
  - For the movement of atoms ...
  - For the movement of galaxies ...
  - For the human issues we are concerned about?

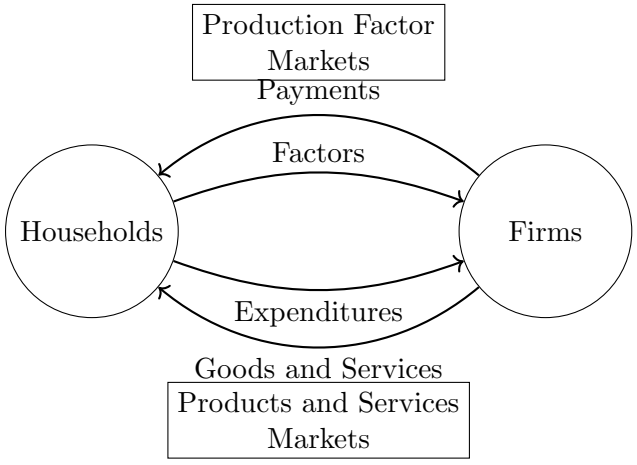
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Slide 7: The Economic System

- Households own factors of production, sell (services of) to firms.
- Households receive payments for provision of factors.
- Households buy goods and services from firms.
- Households expenditures are payments to firms for goods and services.
- What is missing?



Introductory courses in economics typically present a circular model of the economy. In this model, there are two elements, households and firms. These elements interact in two markets, the factor market and the market for products and services. Households own the factors of production, which includes owning the firms. These factors are labour time, financial assets, entrepreneurial abilities, and land. Labour time is sold to firms for wages, financial assets are provided to firms in exchange for dividends and interest payments and entrepreneurial abilities are paid for as profits. Households also own land, which they rent to the firms.

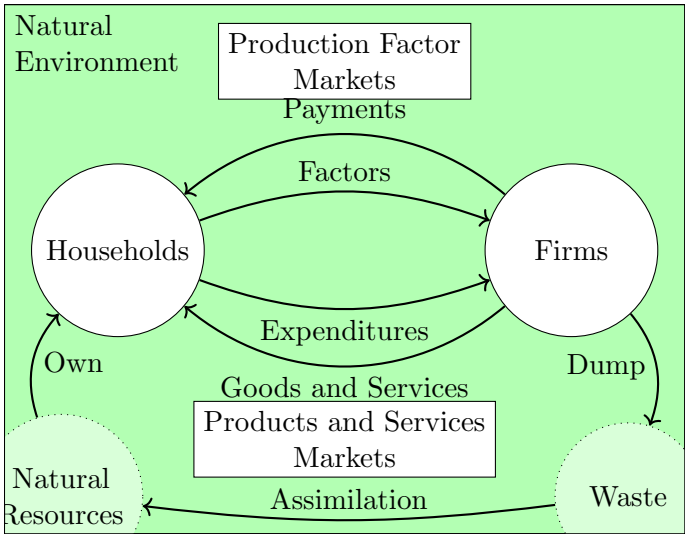
The firms take the factors of production and transform them into the goods and services that households consume. Households purchase these goods and services in the market for products and services. Household expenditures are the revenues that the firm collects.

Economics is sometimes defined as the discipline that studies how households and firms make decisions in the face of scarcity. Many of the problems that are the focus of economic research are where that scarcity is financial. The contemporary (in 2026) concerns about the cost of living, the so called housing crisis, and concerns about AI replacing workers are concerns about financial scarcity. Many people's income is insufficient to purchase the things they need - daily necessities and a decent place to live - and/or that that income is precarious.

There is a substantial body of work - academic and otherwise - that speaks about this precarity leading people to focus on their immediate needs and therefore put less concern on the environment. Opinion poles and recent election results bear this out. There are important political dimensions to this. Power relationships and how they shape the political space and the decisions made in that space are a very important part of the system. However, for the time being, the aim here is to develop some of the tools that have been developed and used in the economics discipline to understand sustainability challenges.

Slide 8: The Ecological Economic System

- Households own natural resources (Land).
- Households rent land to firms, that use natural resources.
- Firms (and households) produce waste, dumping it into the environment.
- What would make this sustainable?
- Sustainable if all waste assimilated into natural resources.



There are many critics, both within the economics discipline and outside of it, who emphasize that the circular model of the economy is incomplete. The flow of payments from firms to households that enables households to purchase the goods and services firms produce is fair to consider a closed system. In fact, this is a central relationship that macroeconomists use when tracking the performance of the economy as a whole. However, it does leave the illusion that the goods and services that households purchase and consume are transformed by households back into factors of production that the firms use.

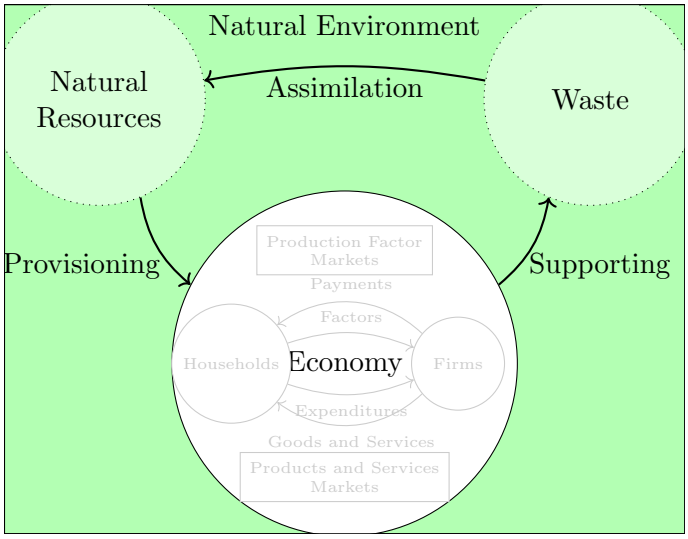
It is often suggested by these critics that students of economics need to be taught right from the start that the economy is embedded within the natural environment. From a systems perspective, the economy is a subsystem of the natural environment.

If the scale of analysis and focus of the question is the relationship between the economy and the natural environment, then everything that goes on inside the economy can just be left there. The economy would then be like any other agent in the natural environment, an organism that takes in resources and expels waste. The ecosystem is sustainable when all the waste produced by this economy subsystem is assimilated by processes in the environment that convert it back into natural resources.

How do we get to a situation where these flows are stable, where none of the elements are growing at the expense of other elements.

Slide 9: The Ecological System

- The economy as an element of the ecological system.
- The natural environment provides resources needed by the economy.
- The natural environment supports the economy by receiving waste.
- The system is sustainable if the flow of waste is fully assimilated back into natural resources.



In both the natural and social systems, researchers try to identify patterns in the systems that they are studying. Those patterns emerge from the interactions between the elements that make up the system. When systems are large and complex, the discipline typically finds ways to describe the patterns without examining the detail of the interacting entities that leads to the pattern.

Cosmologists studying the movement of galaxies do not spend much time considering the movement of individual hydrogen atoms. However, it is the movement of those individual hydrogen atoms, under the influence of gravity, that leads to the formation of stars, and that leads to stars spinning together in galaxies.

Often, the disciplines have observed or thought deeply about the entities that they are interested in, and arrived at relationships - often mathematical - that represent the patterns that they observe.

In chemistry, the relationship $PV = nRT$ is known at the Ideal Gas Law. It is the relationship between pressure (P), volume (V) and temperature (T). The amount of gas in the volume is (n) in units of moles, and R is the ideal gas constant, expressed the the appropriate units. A mole is Avagadro's number of atoms. This relationship expresses how temperature, which is a measure of how much energy atoms have, and the pressure that they are exerting within a particular volume. The physics of the situation is that the atoms are racing around, their speed reflecting the temperature, and smashing into the sides of the container that confines them. The more energy they have, the faster they are moving, and the harder they collide with the chamber. However, the equation is not representing the movement of those individual atoms. Rather, it is expressing this effect in a way that is relevant for things like the operation of a steam engine.

In the last few decades, computers have become powerful enough for us to represent the details of relationships like this from the basic building blocks. Agent based models are a class of such tools. One such agent based modeling environment we will use from time to time is NetLogo (https://www.netlogo.org/).

Slide 10: Agent Based Models

- Agent Based Models (ABMs) that interact virtual agents in virtual environments, ot observe how the system evolves.
- Agents can range from individual atoms to galaxies.
- Agents have often simple rules of behavior that they are programmed to follow.
- An ABM allows experiments with these agents, observing how the system that contains them evolves as a result of their interactions with other agents and the environment.
- ABMs allow properties of the system to emerge, properties that are not inherent to the agents that make up the system.
- ***emergence*** is an important characteristic of complex systems.

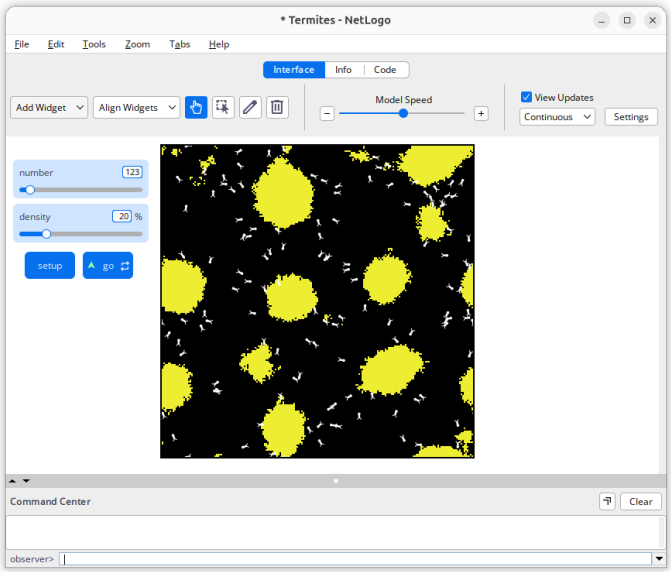
Building an agent based model and implementing it is a significant computational exercise. However, the careful thinking process that goes into the design of the model is valuable, even if one never codes it up (with or without the help of AI). This is because it forces us to think systematically about the details of how people respond to each other and to their environment.

Slide 11: Agent Based Thinking

- To build an ABM need:
  1. A collection of agents with clearly defined behaviors.
     - How they respond to other agents.
     - How they respond to the environment.
     - How they evolve over time.
  2. A way for the agents to interact with each other and the environment.
     - Which agents interact.
     - How interactions change.
  3. A process for how the environment changes.
     - How the environment responds to agents.
     - How the environment evolves over time.
- Such careful thinking - looking at the trees - can illuminate features of the system that are not obvious when only looking at the forest.

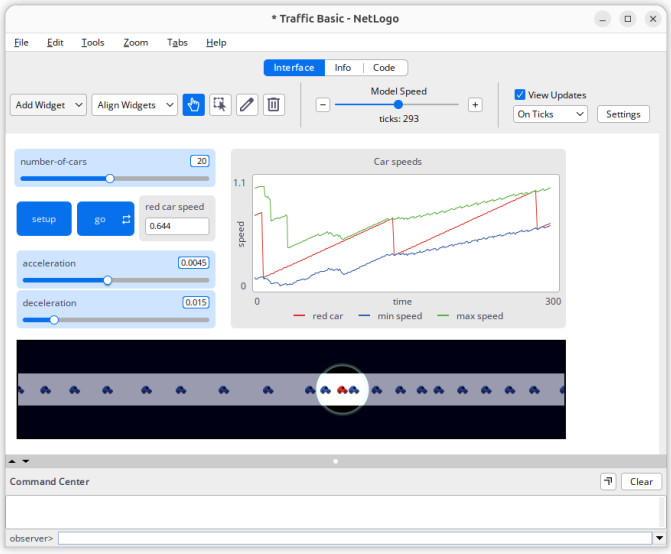
Slide 12: Termites and Wood Chips

- Termites
  1. Move randomly around landscape. Can carry one wood chip.
  2. If not carrying, encounter a wood chip, pick it up.
  3. If carrying, encounter a wood chip, put carried chip down in adjacent empty spot.
- Environment initialized with random distribution of termites and wood chips.
- Each time step, termites move one space on landscape.
- Implemented in NetLogo (https://www.netlogo.org/)



Slide 13: Traffic

- Cars
  1. Accelerate if space ahead, until speed limit.
  2. If car ahead, slow to a bit less than speed of car ahead.
- Environment a linear one lane road.
- Each time step, cars move forward, adjusting speed based on other cars.
- Implemented in NetLogo (https://www.netlogo.org/)



Slide 14: Summary

- A system is a collection of connected elements.
- We inhabit a complex, adaptive social ecological system.
- How a system is analyzed depends on a chosen boundary, spatial and temporal scale.
- System models used in economics often focus on elements connected by financial flows.
- Economic system models can be coupled with the natural system to explore economy environment relationships.
- Careful, detailed description of the elements of a system and their relationships can clarify system dynamics.
- Tools such as agent based models can reveal emergent properties of systems that do not obviously follow from characteristics of agents and their relationships.

Part II

Foundational Systems

Chapter 3

Social Ecological Systems #1

When people who are concerned about the environment think of economics, they often conclude that it ignores the environment. The way introductory economics is taught often reinforces this, as if the environment is mentioned at all, it is a small topic, externalities.

There is a practical reason for this. During a presidential debate, former US president Bill Clinton said "It's the economy, stupid." [CONFIRM QUOTE]. He was emphasizing that what people are most worried about is their own situation. In a modern economy, your situation depends on your income and what it costs you to live. The environment is important, but is secondary to being able to put food on the table tonight, and having a roof under which that table is placed.

You are in this course because you are concerned about the environment, about sustainability. Therefore, lets start by building our understanding of sustainability economics from that environment.

Slide 15: The Fish Pond

- The Question: How does the population of fish in a pond change over time?
  - What time step?
  - One minute? - population changes small, unless major event.
  - One century? - population change in relation to long term events.
  - One year? - scale that matches seasons.
- The elements of a fish pond.
  - The fish (each fish, or fish as a population - biomass?).
  - The water (each molecule, or as a medium for the fish?).
  - The food (each food item - e.g phytoplankton, insects, other fish - or as a population - biomass?).
- The connections within the fish pond system.
  - Fish encountering food and eating.
  - Fish health and survival affected by state of the water.
  - Fish encounter each other and reproduce.

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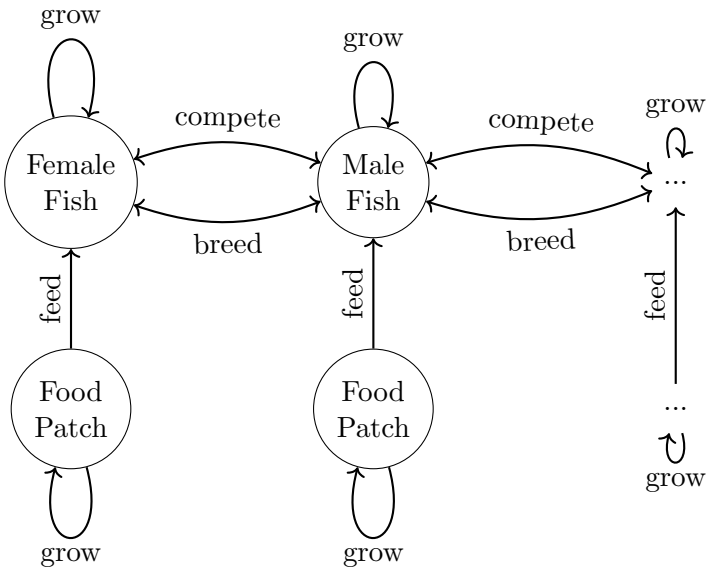
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Fish example: https://virtualbiologylab.org/ModelsHTML5/PopGenFishbowl/PopGenFishbowl.html

Slide 16: The Fish Pond



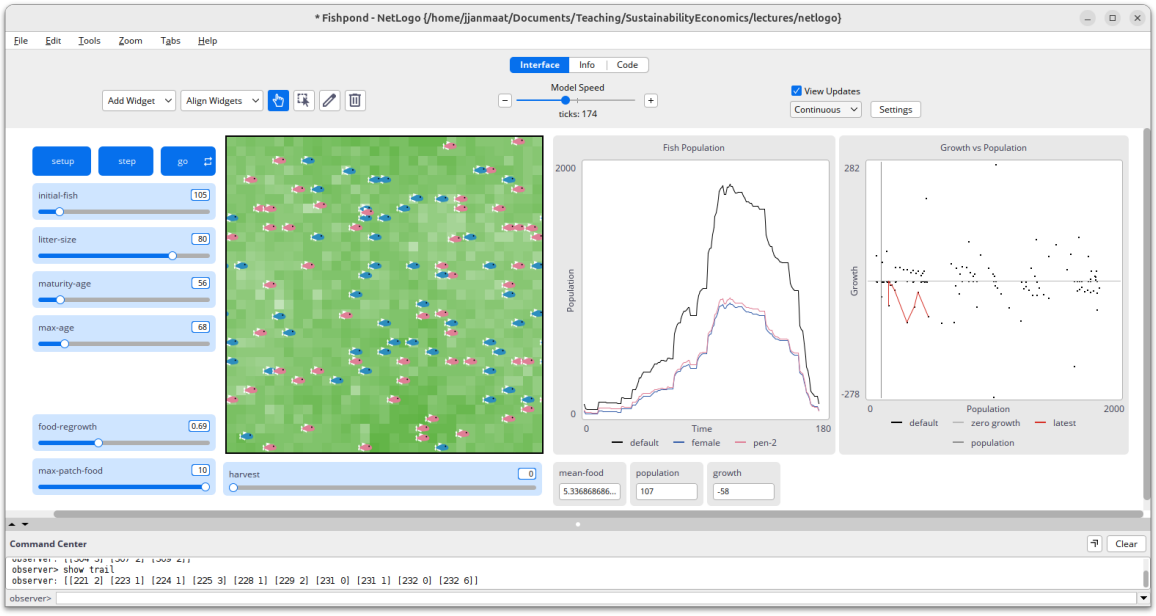
Slide 17: The Fish Pond

- What question(s) are we trying to answer with a model?
- What features of the elements (fish and food patches) and their interactions will lead to a sustainable system.
  - Sustainable = does not collapse to a system without fish.
- What level of harvest is sustainable?
  - Sustainable = harvest level that can continue without collapsing system to one without fish.
- How much detail do we need to model about the ecological system to identify the sustainable harvest level?
  - Can elements of fish pond system be treated together as one element of a social-ecological system?
  - Is the internal workings of the fish pond subsystem necessary to identify sustainable harvest levels?

Slide 18: The Fish Pond

- A NetLogo model.
- Fish (agents) swim around in pond with patches (environment) where they feed.
  - Fish: age to breeding, lifespan, how much food, impact of competition, breeding behavior, movement, ...
  - Food patch: how much food, growth rate of food.
- Lots of details that can be considered.
- Do we loose sight of the forest by focusing too much on the trees?

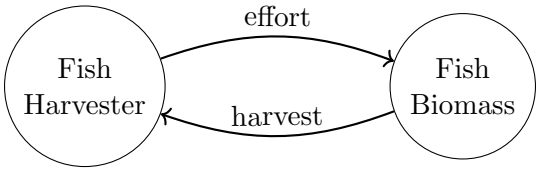
Slide 19: The Fish Pond



Slide 20: Fishing in the Pond

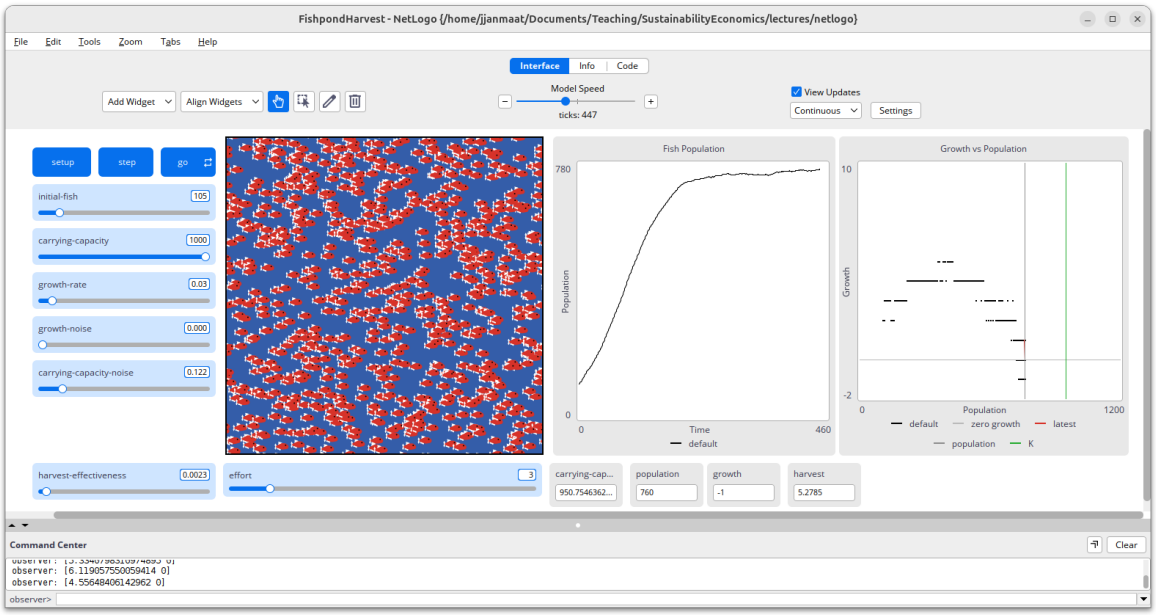
- Fish harvester chooses how much effort (e.g. time) to spend fishing.
- Most processes in pond not affected by amount of fishing effort.
- Effects of harvesting:
  - Reduces population of fish.
  - Reduces fish on fish competition - increasing growth.
  - Increases success of reproduction.
  - May force population so low that collapses.
- Harvest effort:
  - With same effort, more fish caught if more fish in pond.

Slide 21: Fishing in the Pond



- What effort level will result in a sustainable harvest?

Slide 22: Fishing in the Pond



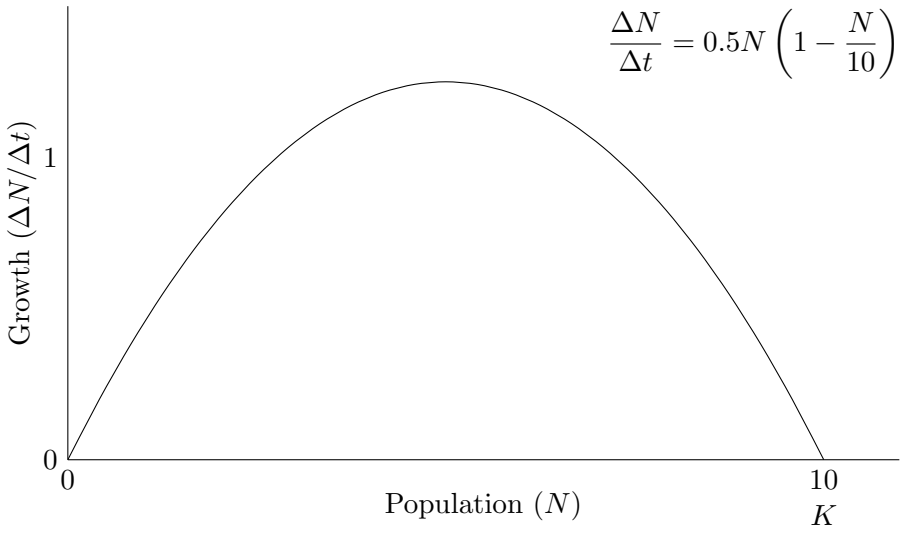
Slide 23: Fishy Math

- Fish population modeled with logistic growth curve.

$$\frac{\Delta N}{\Delta t} = rN\left(1 - \frac{N}{K}\right) = \frac{\partial N}{\partial t} = \dot{N}$$

  - ΔN is change in fish population during time interval Δt.
  - N is fish population.
  - r is rate of growth of fish population.
  - K is carrying capacity of the ecosystem represented by the model.

Slide 24: Fishy Math



Slide 25: Fish Production Function

- In economics, a ***technology*** refers to a process that transforms ***inputs*** into ***outputs***.
- A function that describes this process is called a ***production function***.
- In this example, effort is the input, and harvest is the output.

$$H = qEN$$

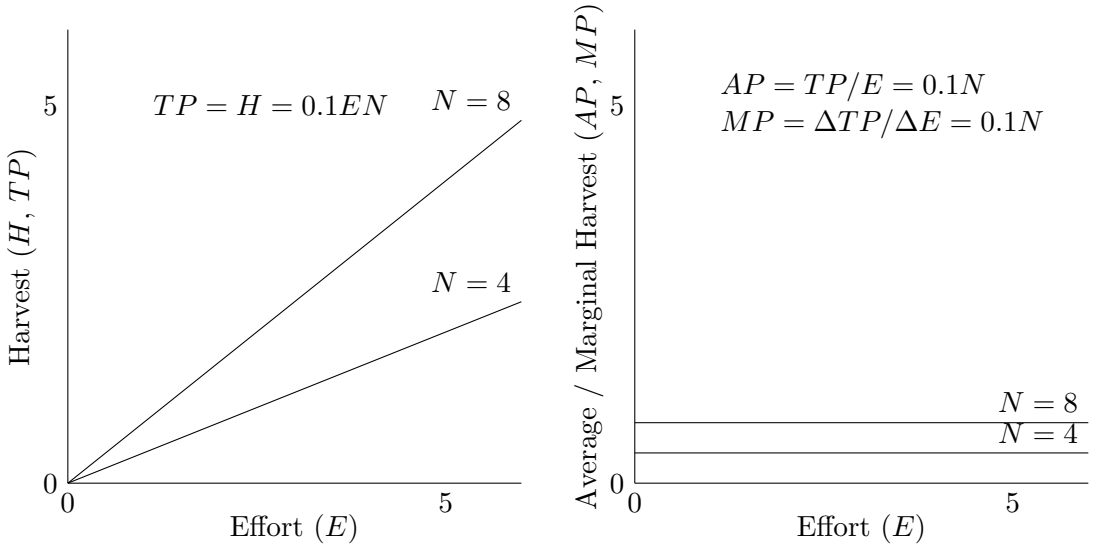
  - H is the harvest.
  - E is the effort.
  - N is the population of fish.
  - q measures how catchable the fish are.
- Note that the productivity of fishing effort depends on the size of fish population (the stock).
- A unit of effort E harvests a fraction q of the stock N.

Slide 26: Fish Production Function

- There are three measures that are commonly used in economics to describe aspects of the production process.

  1. ***Total Product (TP)*** is the total output at a particular input level X.
  2. ***Average Product (AP)*** is the total output at a particular level of input, divided by that level of input.
  $$AP = \frac{TP}{X}$$
  3. ***Marginal Product (MP)*** is, at a particular level of an input, the amount by which output will increase for a small increase in that input.
  $$MP = \frac{\Delta TP}{\Delta X} = \frac{\partial TP}{\partial X}$$
     - The marginal product is the slope of the total product at X.

Slide 27: Fish Production Function



The average and marginal product are the same for this production function. Given a particular fish population, each unit of effort catches the same number of fish. This would stop when there are no more fish left. For the range over which effort is shown in the figure, this does not happen.

Suppose the harvester is deciding how much effort to exert this period, which we will call a year. It is important to recognize that the choice of effort this year affects how many fish will be available to be caught next year. The fish that are left behind reproduce, and this reproduction adds to the supply of fish that can be caught next year.

We can throw some numbers at the example to make it a bit more concrete.

Slide 28: Fish Production Function

- Increasing effort increases harvest.
- Larger harvest means less fish left behind to reproduce.
- There is an ***opportunity cost*** to catching more fish today in terms of the lost contribution to growth of the fish stock in the future.

In the table below, the size of the fish stock, the amount harvested and the growth of the fish stock following harvest is shown for the first ten years, for effort levels $E = 2$ and $E = 3$.

The sequence within a year is that first fish reproduce and then the harvest is taken. What is left of the fish stock is the growth less the amount harvested.

The numbers in the table show a clear trend. For these two effort levels, the stock of fish and the harvest decrease over time. For the lower effort, the beginning harvest is lower. However, if this lower effort is maintained, then the lower harvest leads to a slower decline in the fish stock, and by the end of the ten years, the lower effort is resulting in a higher harvest than for the higher effort level.

If you look closely at the numbers, you can see that the differences from one year to the next are getting smaller. In each case, with the constant effort level, the system is converging on a ***steady state***. At the steady state, the size of the stock, the amount that is harvested, and the amount by which the fish stock grows are constant. In fact, the growth exactly matches the amount harvested. Each of these steady states is in effect sustainable. Which one do we like better?

Slide 29: Fish Production Function

$$H = 0.1EN \qquad \frac{\Delta N}{\Delta t} = 0.5N\left(1 - \frac{N}{10}\right) - H$$

t	N	E	H	ΔN	N	E	H	ΔN
1	10.00	2	2.00	0.00	10.00	4	4.00	0.00
2	8.00	2	1.60	0.80	6.00	4	2.40	1.20
3	7.20	2	1.44	1.01	4.80	4	1.92	1.25
4	6.77	2	1.35	1.09	4.13	4	1.65	1.21
5	6.51	2	1.30	1.14	3.69	4	1.48	1.16
6	6.34	2	1.27	1.16	3.38	4	1.35	1.12
7	6.23	2	1.25	1.17	3.14	4	1.26	1.08
8	6.16	2	1.23	1.18	2.96	4	1.19	1.04
9	6.11	2	1.22	1.19	2.82	4	1.13	1.01
10	6.08	2	1.22	1.29	2.71	4	1.08	0.99

The following graphs show the progression of the stock as the continuing harvests change the size of the stock and the growth.

The system starts at the carrying capacity K. At the carrying capacity, the stock is steady. The new fish added to the population, the recruitment, is exactly matched by the deaths. There is no net growth of the stock.

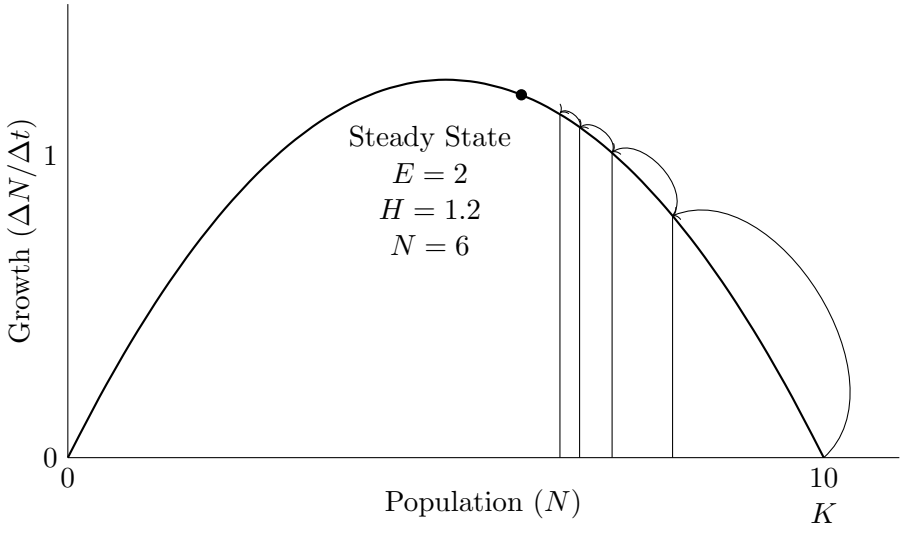
The first harvest reduces the stock by the amount of the harvest. When the effort is lower, this harvest is lower. As the stock is now below the carrying capacity, it has room to grow. That growth is represented by the height of the curve.

The second harvest captures a quantity of fish equal to all the growth and a bit more, reducing the stock further. This effect is larger for the case with the larger effort. With the stock reduced further, there is more capacity for growth.

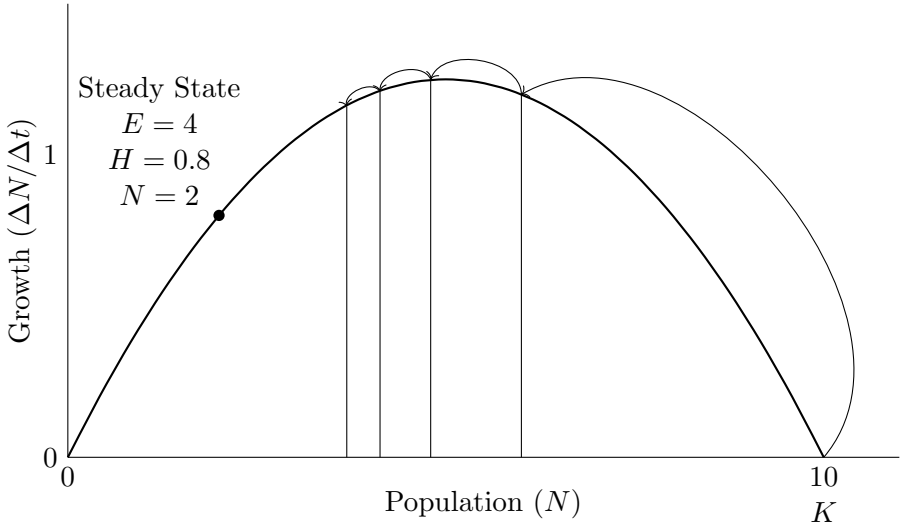
In the figures, things begin to diverge with the third harvest. When the effort is low, the change in the stock after the harvest leads to an increase in the growth of the stock. However, when the effort is high, the stock has been depleted so much that the further depletion results in a lower growth rate. We have moved the stock to a place where there is little competition for food, but it takes a lot of effort to find others to mate with.

If we continue with the same effort levels, the systems will converge on a steady state. The steady state is the level of the stock such that, given the effort level, that stock does not change any further. In these examples, the higher effort level leads to a lower steady state stock and a lower steady state harvest than for the lower effort level.

Slide 30: Fish Production Function



Slide 31: Fish Production Function



Slide 32: Fishing Sustainably

- Starting from fish population $N = 10$:
- Effort $E = 4$ leads to:
  - An immediate harvest of $H = 4$
  - A steady state with $N = 2$ and $H = 0.8$
- Effort $E = 2$ leads to:
  - An immediate harvest of $H = 2$
  - A steady state with $N = 6$ and $H = 1.2$
- Steady states are "sustainable", in that the activities can continue indefinitely.
- Is one steady state better?
- Can it be better to take big harvests in the early years, ending up with a lower sustainable harvest?

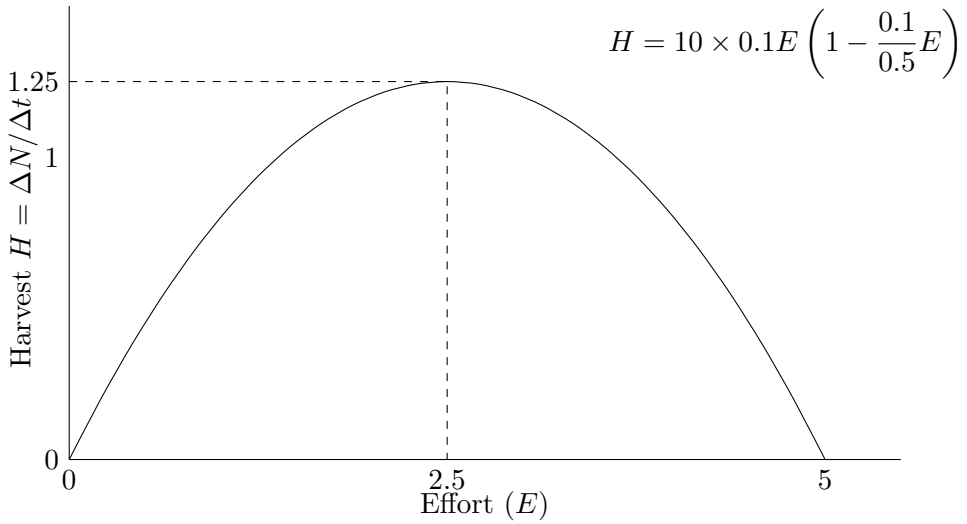
Each effort level corresponds to a unique steady state, or sustainable, stock and harvest level. These combinations are shown in the figure. Notice that the curve has the same shape as the stock and growth curve. However, the horizontal axis represents the effort devoted to fishing, and the vertical axis the harvest. The higher the effort level, the lower the corresponding steady state stock. The stock will be at its carrying capacity when there is no effort, when no fish are being harvested. With high enough effort, all the fish will be caught and the stock will be driven to zero.

Is an effort level that drives the stock to zero sustainable? Seems strange to consider. However, if the stock is zero and therefore the harvest is zero, this will continue to be the case into the future.

In a strange way, a fish stock that has been driven to extinction is now sustainable, as it is not changing any more.

This highlights that when we talk about sustainability, we are not just talking about a technical state of a system. Sustainability as a concept, in most of the ways that we use the idea, is not the same thing as a steady state. There is something more to it.

Slide 33: Fishing Sustainably



Slide 34: Fishing Sustainably

- Each effort level corresponds with a steady state "sustainable" stock size and harvest level.
- There is one effort level that corresponds to the maximum sustainable harvest.
  - In the fisheries literature, this is the ***maximum sustainable yield (MSY)***.
- When is it best to manage a resource to achieve the maximum sustainable yield?
- What is the ***function*** of this harvester and fish pond system?
- What would it mean for this harvester and fish pond system to be ***resilient***?

Slide 35: Summary

- Biological resources like fish typically grow toward a carrying capacity.
- A fish pond can be thought of as a system. Fish compete for food, reproduce, and die.
- A simple agent based model can demonstrate the convergence of a fish population to its carrying capacity or collapse, depending on the characteristics of the system.
- Harvesting fish from a fish pond affects the growth of the fish population.
- Harvesting another fish today has an ***opportunity cost*** in reduced fish population to harvest tomorrow.
- Different effort levels correspond to different sustainable fish harvest levels, with the maximum harvest being the ***Maximum Sustainable Yield***.
- The relationship between harvest and effort is the ***Production Function***, which can be described with the ***Total Product***, ***Average Product***, and ***Marginal Product***.

Chapter 4

Hermit Economics

4.1 Learning Objectives

- Explain the meaning of opportunity cost.
- Demonstrate an understanding of the production function and the production possibilities frontier.
- Explain the role of diminishing marginal product.

4.2 The Hermit - Part One

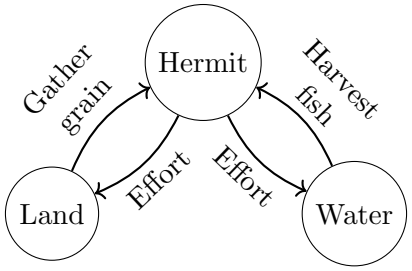
A hermit is someone who has chosen to live apart from society. In the most extreme case, they have no contact with any other person and are completely self sufficient. From a system thinking perspective, the hermit would be one element of the system, with the other elements being part of the ecological side of the hermit-nature social ecological system.

Slide 36: Hermit System

- Hermit,
  - The social element of the social ecological system.
  - Contributes labour and consumes the fruits of that labour.
- Field,
  - One ecological element of the system.
  - Produces grain, with labour as the main input.
- Lake.
  - One ecological element of the system.
  - Produces fish, with fishing effort (time spent fishing) as the main input.

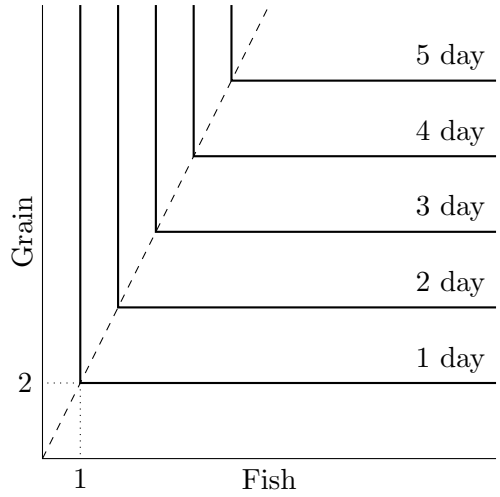
The hermit can apply their time either to labour in the field, growing grain, or effort at the lake to catch fish. The more labour applied in the field, the more grain that the hermit harvests. Likewise, the more effort devoted to fishing, the more fish that will be caught.

Slide 37: Hermit System



Slide 38: Human Ecology

- Humans need food.
- Carbohydrates, fat and protein.
- Grain high in carbohydrates
- Fish provide fat and protein.
- Assume need fixed amounts of each.



Humans need food to survive. The basic components of that food are carbohydrates, fat and protein. There are various specific nutrients and micronutrients we need. So long as the fish and grain in our example provide sufficient quantities of these, we can focus on the balance of the three main nutrients. For our model, we assume that the grain provides the carbohydrate requirements and the fish provide the fat and protein.

One way to think about this is in terms of the number of days that a person has enough grain and fish to survive. We can show this graphically using a contour map. The dimensions of the map are the amount of fish and the amount of grain that the person has in their possesses. Each contour line represents a day of survival.

The contours are 'L' shaped because survival for a day depends on consuming a minimum amount of each food source. Consider the contour representing one day of survival. The point

where $q_F = 1$ and $q_G = 2$ is, in this example, the minimum amounts of fish and grain that are needed to survive one day. If the person has more fish, but only the two units of grain, they will not survive more than one day. Thus, with two units of grain, the contour for all larger quantities of fish always represents the same one day of survival.

The same logic applies when the person has one unit of fish. For any amount of grain that is more than two, the person cannot survive any additional days. Therefore the contour is vertical at the minimum amount of fish needed to survive one day.

For the mathematically inclined, the function for the contours is

$$d(q_F, q_G) = \min\left(\frac{q_F}{\alpha_F}, \frac{q_G}{\alpha_G}\right) \tag{4.1}$$

In this equation, α_F and α_G are, respectively, the amounts of fish and grain that the person needs to consume per day to survive that day. One can imagine that these amounts can differ by person. One can also imagine that if there is a change in technology that translates fish and grain into nutrients could also change this relationship. For example, a food preparation method that wastes less would reduce the amount needed per day of that food item.

What have we not considered? What would change if fish and grain decay? The fact that the kinks in the figure follow a straight line, and the spacing of the contours is always the same assumes that there is no decay. We know that grain can be stored far longer than raw fish. If the fish is not processed into something that can be stored - dried for example - then how would the figure change? What would happen if a share of the fish is lost each day? What would happen if no fish can be consumed after it is five days old?

Another thing to consider is that while grain has far more carbohydrates than it does have fat and protein, it does have some. How would this change the graph? In those cases, it will not have the tight kink. Instead, it will have a bit of a curve, since there is some ability to substitute more grain for less fish.

Slide 39: Production Functions

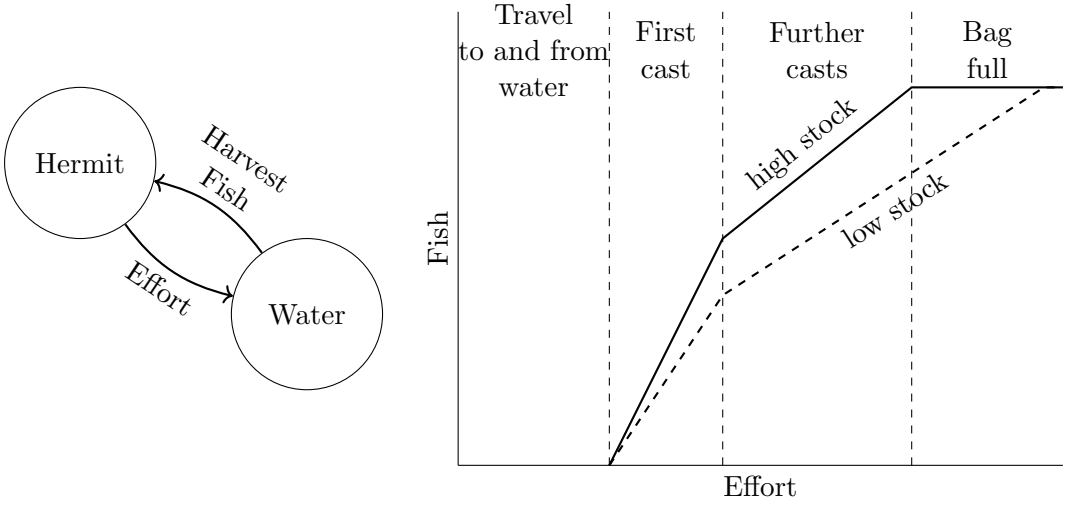
- A *technology* is a way that inputs are converted to outputs.
- A *production function* is a way of representing the technology.
  - The production function is a simplification.
  - The technology behaves the way that physics, chemistry, etc. require it to.
  - All models are wrong, some are useful.
- Three measures are commonly used to describe production at each input level.
  - *Total Product* is the total output produced given a particular input level.
  - *Average Product* is the total product divided by the input level.
  - *Marginal Product* is how much the total output changes if the level of input is changed by a small amount.

Slide 40: The Fish Subsystem

- Time path:
  1. Travel from village to water
  2. Cast net
  3. Collect fish from net until can't carry any more
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

If we consider the fish subsystem in relation to the village, we need to add a couple of elements that are not there when we just consider the relationship between effort and fish caught. The fish harvester also needs to spend time traveling between the village and the water where fishing is done.

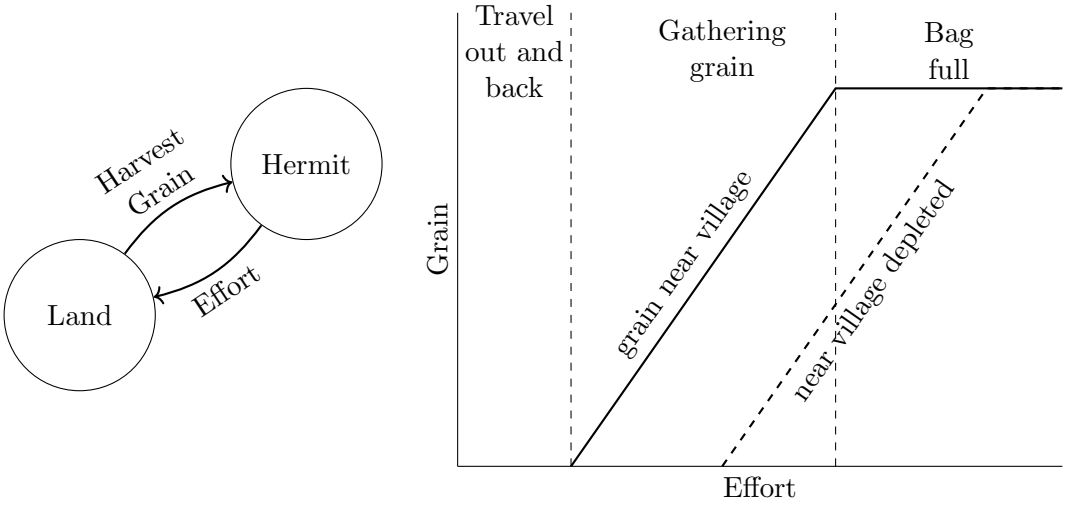
Slide 41: The Fish Subsystem



Slide 42: The Grain Subsystem

- Time path:
  1. Travel from village to closest patch of land with grain.
  2. Harvest grain from patch.
  3. Move to next patch and repeat until grain bucket full.
  4. Travel back to village
  5. Exchange fish for grain
- Each step of time path requires effort.

Slide 43: The Grain Subsystem



Slide 44: Total, Average and Marginal Product

- Total Product (TP) and Average Product (AP) are based on a level of input.

$$AP = TP/E$$

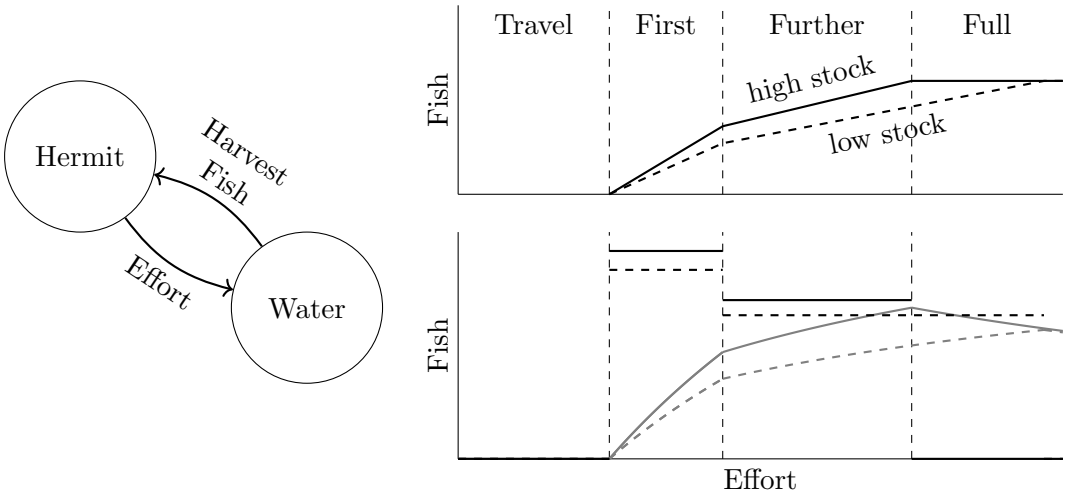
  - E can be any input, not just effort.

- Marginal Product (MP) is based on a small change in in the level of an input, around a specific level of that input.

$$MP = \Delta TP/\Delta E = \partial TP/\partial E$$

  - The marginal product measures how much extra output would be produced if a small amount of extra input is used.
  - It is the slope of the production function at the input level.
  - If production function is curved, slope of tangency line.

Slide 45: The Fish Subsystem

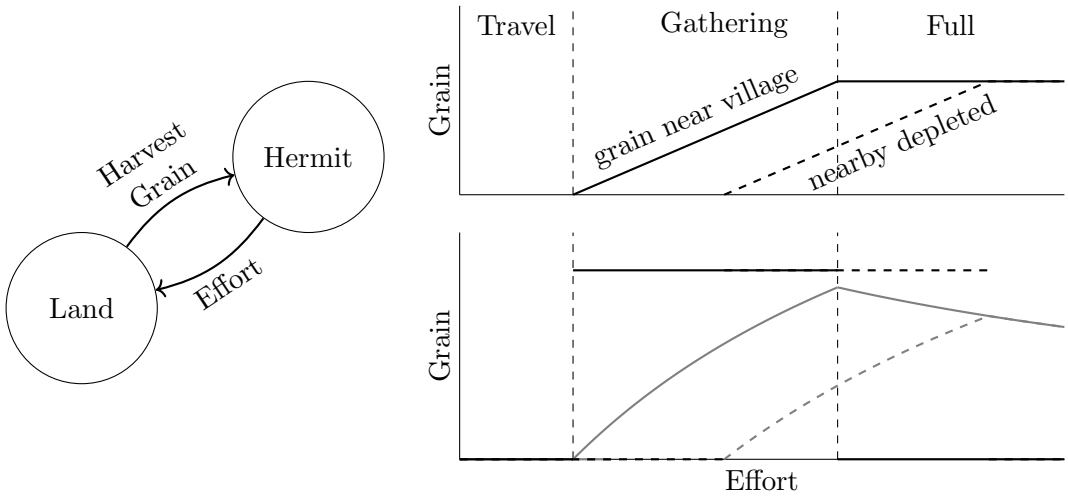


The average product for the fish production function is zero until the level of effort is reached where the effort is sufficient for travel to and from the hermit's home. After this point it climbs, at a slowly decreasing rate. The curvature is subtle in the graph, but it is there. Once the first net cast is hauled in, the later casts catch less fish. The rate of increase of the average product slows. Once the fish basket is full, there are no further fish caught. From this level, the average product is downward sloping, with the slope becoming less steep as effort continues to increase.

The marginal product is also zero until the first net cast. Through the first net cast, it is at its highest. When fish are being pulled into the basked from the second cast, the extra fish per unit of effort is lower. Once the basket is full, extra effort does not add any more fish to the basket, so the marginal product is again zero.

Notice that when the average product is increasing, the marginal product is above the average product, and when the marginal product is decreasing, it is below the average product. This is a general result.

Slide 46: The Grain Subsystem



The average and marginal product curves for the grain gathering are similar to those for the fish harvesting. In both cases, the marginal and average product are both zero until the travel requirements are satisfied. After that, the average product increases until the bucket is full. Since the slope of the production function is constant, the marginal product is also constant. Notice that with the difference between the two production functions being only the effort needed to get to and then back from the patches with grain, depleting the patches near the village does not change the marginal product. It does lower the average product, since the total effort needed for any amount of grain gathered is higher.

Our hermit can divide their effort between gathering grain or harvesting fish (or staying at home and resting). We can plot these combinations of fish and grain in what in economics is call a ***production possibilities frontier (PPF)***. It is a frontier because it is what could be produced if all effort is used. In the figure, the combination marked E_1 is the amount of production of fish and grain that can occur if there is only enough effort available to travel between the harvest and gathering sites and the hermit's home.

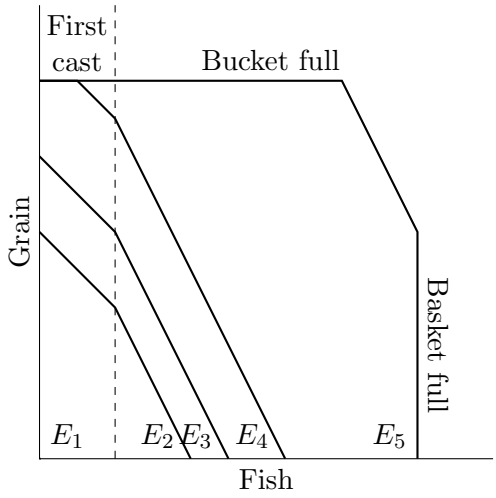
If the hermit has sufficient food to allow them to survive beyond the travel time, then we can examine the combinations of fish and grain that the hermit can produce by dividing their effort in different ways between fishing and gathering. For the curves represented by E_2 and E_3, if the effort is reallocated from harvesting fish to gathering grain, the amount of fish harvested decreases and the amount of grain harvested increases.

With the PPF labeled E_4, the limit of what the fish basket can hold has been reached. In effect, the hermit 'throws away' part of the total effort, as it is dedicating it to catching fish, which cannot be take back home. Since the first cast is the most productive, the hermit is in fact throwing away the most productive effort when running up against this constraint. If this seems strange, think of the order of events. The hermit cannot devote effort to later catches until they have made the first cast.

The PPF labeled E_5 corresponds to an effort level where one of the grain bucket or fish basket is full for most of the effort divisions. There is in effect a lot of wasted effort here, as it is being applied to activities which cannot result in any more grain or fish being brought back home. We might think of this wasted effort as leisure. If the hermit has enough food to sustain themselves for this many days, maybe it would be better for them to stay at home, enjoying themselves while eating the stockpiled food, heading out when they have enough food left to fill their basket or bucket and bring it back home.

Slide 47: Production Possibilities

- Split total effort between fish and grain.
- $E_5 > E_4 > E_3 > E_2 > E_1$
- At E_1, no fish caught or grain gathered.
- At E_5, most effort not used, as bucket or basket full.



The PPF maps the different combinations of fish and grain that can be produced for a particular level of effort. It shows the impact of reallocating effort from the production of fish to the production of grain. Moving between points along the PPF involves reducing the amount of one good produced and increasing the amount of the other produced.

Opportunity cost is defined as *the value of the next best alternative foregone*. The production possibilities frontier is one way to illustrate opportunity cost. In the figure, three changes in the amount of grain are produced, for the PPF that was labeled E_4 in the previous figure.

If fish production is increased by ΔF_1, there is no reduction in grain production. For this change in fish, the grain bucket is full and the effort beyond filling the grain bucket is not adding anything to the bucket. In this situation, moving effort to harvesting fish has no impact on the grain gathered. The opportunity cost of this increase in fish harvested is zero grain gathered.

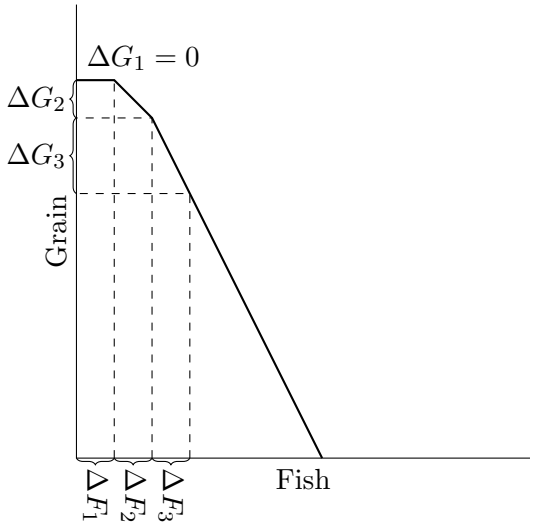
For the increase in fish represented by ΔF_2, the amount of grain harvested falls by ΔG_2. The effort being applied to fishing here is part of the first cast, when the effort is most effective. When more effort is put to fish harvesting, adding ΔF_3, we have moved into the further casts, where less fish have had time to move into the area where the net is cast, and therefore less fish are collected for the same amount of effort. Therefore, there is now a larger amount of effort taken away from gathering grain to realize the increased fish harvest.

This is a result we expect to be generally true. As we take more resources away from producing one good, in this case grain, the resource that we are taking away is more productive. With the

production function, we saw that the marginal product decreased as effort increased. As we add more effort, the contribution of an extra unit of effort to total production falls. If we are moving effort from one good to another, the good that we are taking effort away from is moving to a higher marginal product part of the production function, while for the good that is getting more effort, we are moving to a lower marginal product part of the production function. Taken together, as we move from high grain production and low fish production to high fish production and low grain production, the amount of grain we give up to get an extra unit of fish increases.

Slide 48: Opportunity Cost

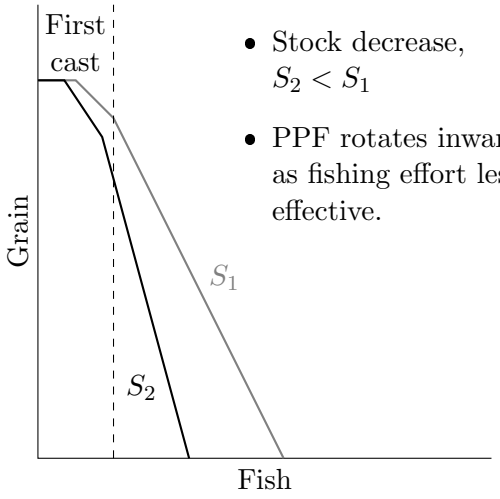
- Increase fish by ΔF_1 for no reduction in grain.
- Increase fish by ΔF_2 for reduction in grain of ΔG_2.
- Increase fish by ΔF_3 for reduction in grain of ΔG_3.



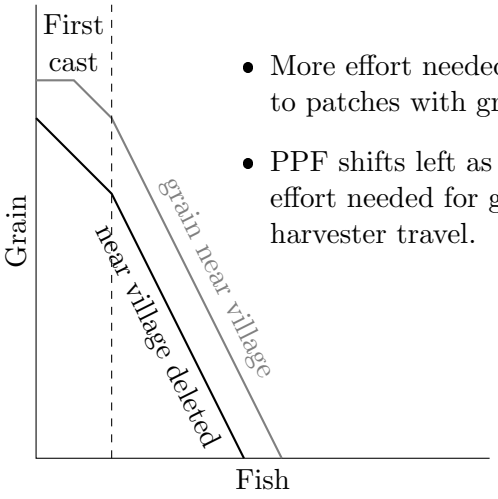
When the production functions were described above, some aspects relevant to sustainability were illustrate that do not show up here. For fishing, if harvesting reduces the fish stock, then the production function slope becomes steeper. It will take more effort, more days spent fishing, to fill up the fish basket. When all the effort is devoted to fishing, the catch will be lower on account of the lower stock. The fish axis intercept for the PPF has shifted inward. The decline in the fish stock does not affect the amount of grain that can be gathered, if all the effort is devoted to gathering grain. Therefore, the intercept of the PPF with the grain axis is unchanged. However, the effort that is made availeble for fishing when the grain bucket is full is less effective with a smaller stock. This effort 'buys' less fish, and therefore the PPF bends down at a lower fish harvest level when the fish scok is less.

When the grain available in patches near the village is depleted, more effort must be spent on travel. This reduces the total effort available for both activities, if we start from the same premise where we only examine how to reallocate effort after enough food is devoted to enabling both the fish harvester and the grain gatherer to travel to and from their respective activity locations. The PPF shifts left, lowering the intersection with the Grain axis. Since the total effort to divide is reduced, there is no longer enough effort to fill the fish basket, even when all the effort is put toward fishing.

Slide 49: Production Possibilities



- Stock decrease, $S_2 < S_1$
- PPF rotates inward as fishing effort less effective.

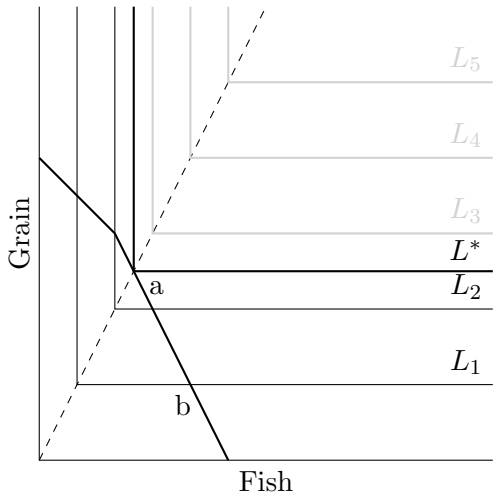


- More effort needed to get to patches with grain.
- PPF shifts left as more effort needed for grain harvester travel.

The production possibilities frontier tells us what combinations of fish and grain can be produced by the villagers, for different levels of effort. The biological reality of our villagers requires that the consume a particular bundle of fish and grain each day. If we put the biology together with the production possibilities frontier, we can see how to allocate the effort to provide the most days of life for the villagers.

Slide 50: Production Possibilities

- Lifespans L_1 and L_2 are attainable.
  - At **b** attain L_1.
  - At **a** attain L^*.
  - Same effort level.
- Lifespans L_3, L_4 and L_5 are not attainable.



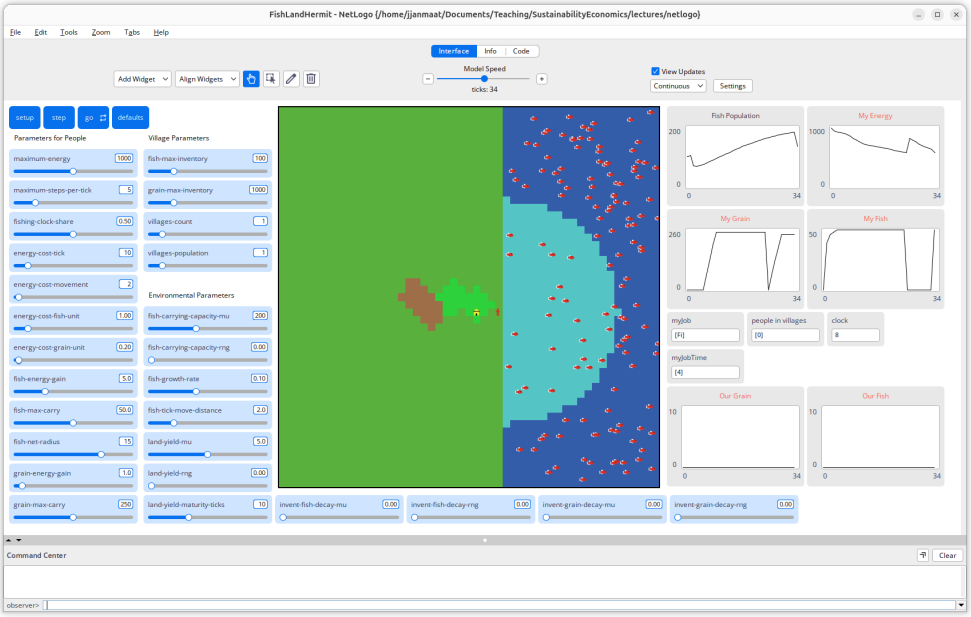
Putting together the human needs and the production possibilities lets us identify something we might see as optimal. It seems reasonable to think that dividing up the resources that the villagers have in such a way as to maximize the number of days that they continue living would be a reasonable objective.

Can we expect this outcome from people who interact independently?

Slide 51: A Hermit Economy Agent Based Model

- The Hermit
  1. Has an energy level that is used up as time advances, with additional energy used for activities.
  2. Can harvest fish or gather grain. Maximum amount of each can carry.
  3. Lives at a location on landscape that must return to at end of each day.
  4. Travels to shore to fish, and back home when time exhausted or basket full.
  5. Travels to nearest grain to gather, continuing moving and gathering till bucket full or time exhausted.
  6. At end of day, consumes fish and grain to increase energy.
- Fish live in pond, population grows as described in previous section.
- Grain
  1. Patches slowly grow grain.
  2. If grain harvested, starts growth cycle.
  3. No grain to harvest till maturity reached.

Slide 52: A Hermit Economy Agent Based Model



Slide 53: Summary

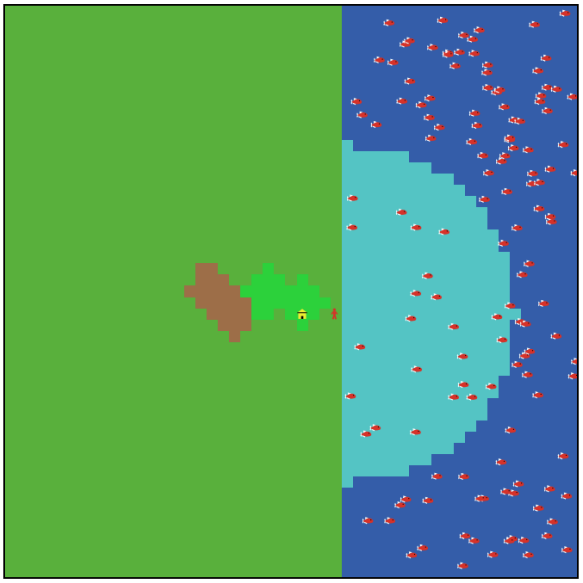
- A simple social ecological system, a hermit economy, is described.
  - The hermit divides their time between harvesting fish and gathering grain.
  - The hermit needs to consume fish and grain in a specific ratio to remain alive.
  - The hermit needs to travel to harvest and to gather.
- Using the fish and grain production functions, it is demonstrated that the marginal product is above the average product when the average product is increasing, and below when the average product is decreasing.
- Combining the production functions, production possibilities frontiers are constructed for different effort levels.
- Opportunity costs are demonstrated using the production possibilities frontier.
- The impact of changing environmental conditions on the production possibilities frontier is shown.
- The division of effort between harvesting and gathering that provides the most additional days of life is identified.

Slide 54: Risk and Uncertainty

- A risky situation is one where the outcome is unknown, but possible outcomes and their odds are.
- An uncertain situation is one where the possible outcomes and/or their odds are not known.
- In our hermit model, fish move randomly.
- How should the hermit respond to this risk?

Slide 55: Fish Harvest Risk

- Fish move randomly around pond.
- Can calculate probability that any location in pond has a fish in it.
- Can therefore calculate probability of any number of fish in area covered by net.
- Can also calculate how many fish likely to move into area for further casts of the net, after first cast.
- Can calculate an expected number of fish caught on any fishing trip.



Slide 56: Expected Value

- ***Expected value*** is a statistical concept, defined as the outcome that is expected to occur for a random event if it is repeated a large number of times.
- A fair coin has equal odds of coming up heads or tails.
- If you are offered \$10 for heads on a coin toss, \$0 on tails, what is your expected winnings?
- The expected value is based on what the outcome would be if this is repeated a large number of times.
- Half of the time you win \$10, half of the time you win nothing. The expected value is \$5.

$$E(\text{coin toss}) = \frac{1}{2} \times \$10 + \frac{1}{2} \times \$0 = \$5$$

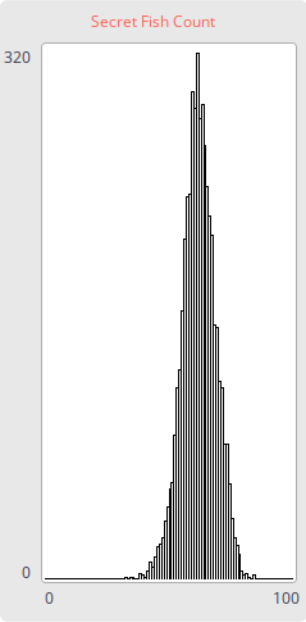
- Need an *exhaustive* set of events and probabilities.

Slide 57: Expected Catch

- For the fish pond, the probability of different numbers of fish in the net area is not the same.
  - Probability of more than carrying capacity is (close to) zero.
  - Probability of no fish is small on the first cast.
  - Probability of lower numbers increases after first catch, falls for higher numbers.
- Graph counts frequency of each number of fish in net span, for 4694 fish random movements around pond.
- Probability is count / frequency.
- For math keeners

$$P(q = j) = \binom{K}{j} \left(\frac{A_n}{A}\right)^j \left(1 - \frac{A_n}{A}\right)^{K-j}$$

q = fish in net, K = total fish in pond, A = area of pond, and A_n = area covered by net.



Slide 58: Expected Catch

- Amount of fish caught depends on fishing effort, fish in pond, and fish that have swam into net area.
- Expected catch - average for many fishing trips - is sum of each possible catch, weighted by probability of that catch.
- More math

$$E(\text{catch}) = \sum_{j=0}^{K} j \times P(j)$$

- Expected value is average catch from many fishing trips.
- On an individual fishing trip, will catch be one of those possible.
  Some of those possible far below expected value!!!
- How should hermit choose to divide effort between grain gathering and fish harvesting?

Slide 59: Expected Value

- Which do you prefer?
  1. \$10,000 for sure (100% probability)
  2. \$50,000 with 10% odds, \$10,000 with 89% odds and \$0 with 1% odds.

Slide 60: Expected Value

- Which do you prefer?
  1. \$10,000 with 89% odds and \$0 with 11% odds.
  2. \$50,000 with 10% odds and \$0 with 90% odds.

Slide 61: Allais Paradox

- Expected values:
  1. Situation #1
     - \$10,000
     - $\$50,000 \times 0.10 + \$10,000 \times 0.89 + \$0 \times 0.01 = \$13,900$
  2. Situation #2
     - $\$10,000 \times 0.89 + \$0 \times 0.11 = \$8,900$
     - $\$50,000 \times 0.10 + \$0 \times 0.90 = \$5,000$
- Majority choose \$10,000 for sure in situation #1 and 10% chance of \$50,000 in situation #2.
- Demonstrates that risk attitudes are not consistent and/or people are not good at evaluating risky situations.

Many of the decisions people make do not have certain outcomes. How would we expect people to act in these situations. We can characterize people as having different risk attitudes. And we might also expect those risk attitudes to differ based on the situation. The Allais paradox highlights how even in simple situations, people's attitudes towards risk can change.

Our hermit needs fish to survive. How at risk is their survival on today's catch? If the hermit is able to store fish, or can survive for several days without fish, then they can manage better than if a poor catch today means they won't survive to tomorrow. If they can survive for several days, then one bad catch can be 'made up for' with a better catch tomorrow. However, if today's catch is essential for survival, then it may be the best survival strategy to put more effort toward fishing, so as to increase the odds of a larger catch.

The less vulnerable someone is to the risky situation, the more likely they are to have a risk neutral attitude. This is even more true if it is possible to ***diversify***. Diversification is the strategy of spreading ones effort across a variety of risky activities. If the likelihood of several bad activities occuring together is low, if the outcomes are not correlated, then the overall outcome is less likely to be bad. It is also less likely to be extremely good, but that is the cost of protecting from the bad outcomes.

Slide 62: Risk Attitudes

- Risk attitudes defined in relation to expected value.
- Would hermit prefer being given their expected catch or going fishing and taking whatever actual catch they realize?
  1. Hermit is ***risk averse*** if would rather take expected catch than go fishing.
  2. Hermit is ***risk loving*** if prefers to go fishing and see what they get, rather than take the expected catch.
  3. Hermit is ***risk neutral*** if they don't care.
- The hermit needs fish to match with grain, in order to get more energy.
- What risk attitude (risk preference) is the hermit likely to have?
- How will the hermit's risk preference likely affect the amount of effort applied to fishing?

The relationship between risk, the hermit's effort choice, and the stock is complicated. We will break it down with a simplification to illustrate the ideas. The flow chart represents choices as circles, with the name in the circle representing who makes a choice there. The rectangles are the outcomes that follow from the sequence of choices that led there.

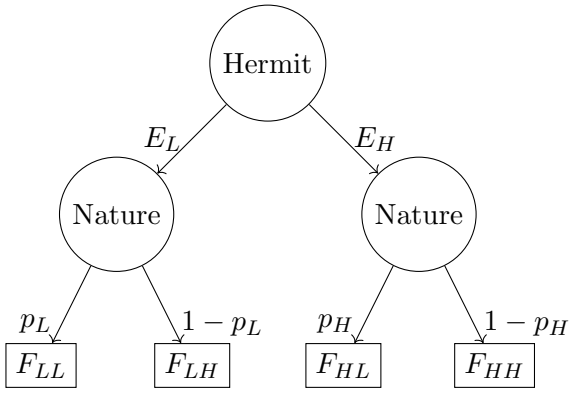
The hermit makes the first choice, that being between a high and a low effort level. Once the effort is chosen, nature gets to make a choice. We do not typically consider nature as choosing, in the same way as a person, who we expect to have some sort of a goal. However, it is convenient to think of nature as a decision maker who 'rolls the dice'. In this case, there are two possible outcomes that nature chooses between, with the probability of choosing the low fish yield being p_L for the low effort branch and p_H for the high effort branch.

Notice that once an effort level is chosen, there are only two possible outcomes. This is shown on the low effort branch by the probability of the high yield being $1 - p_L$, and on the high effort branch, the probability of the high yield being $1 - p_H$. When we have a list with all the possible outcomes and their probabilities, the probabilities must add up to one. Here, on each branch, there are only two possible outcomes, so the probability of one outcome is one less the probability of the other outcome. Of course, no probability can ever be greater than one, as something with a probability of 1 has 100% odds of occurring.

The flowchart represents the decision during one fishing session. The stock of fish in the pond evolves over time. The hermit will keep making this choice, but the size of the fish harvest will change depending on how the stock changes. Increasing effort translates into higher catches on average, and that will on average reduce the stock more. By textiton average refers to the average catch for many fishing sessions where the same effort is applied.

Slide 63: Risk and Sustainability

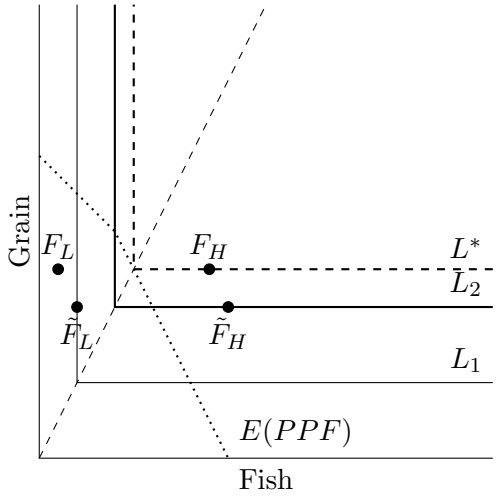
- Hermit chooses effort E_L or E_H.
  $E_H > E_L$
- Nature 'chooses' low (F_{LL}, F_{HL})or high (F_{LH}, F_{HH}) fishing yields.
  $F_{HL} > F_{LL}$
  $F_{HH} > F_{LH}$
- Stock affected by harvest.



We can put together the impact of risk and how the production possiblities frontier relates to the stock size together to show how things can change over time. In the figure on Slide 64

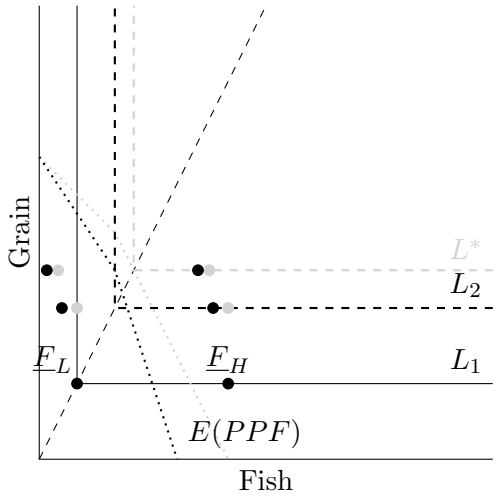
Slide 64: Risk and Sustainability

- $E(PPF)$ expected combinations of fish and grain.
- Effort with expected lifespan L^* gives actual fish harvests of F_L or F_H.
- Effort with expected lifespan L^2 gives actual fish harvests of $\tilde{F}_L$ or $\tilde{F}_H$.
  - Attains L_1 for both harvest realizations..



Slide 65: Risk and Sustainability

- Increased effort reduces fish stock, rotating $E(PPF)$ inward.
- With inward rotation effort with expected lifespan L_2 no longer guarantees lifespan L_1.
- Fishing effort share increases to guarantee L_1.
- To avoid risk of lifespan below L_1, fishing effort share increased, driving stock lower, and hermit just surviving.



Slide 66: Risk and Sustainability

- The reality of fish yield risk creates a reason for the hermit to increase fishing effort. The downside, not enough nutrition to survive, is a more immediate problem than the impact on future yield of reducing the fish stock.
- There is a feedback between effort, catch, and stock that shifts the final outcome to one where the fish stock is lower and the hermit's lifespan 'margin' is smaller.
- Putting more emphasis on the present relative to the future is 'rational' when a bad outcome now can affect whether there even is a future.

A bird in the hand is worth two in the bush.

Slide 67: Uncertainty

- If the hermit is around today, and has access to media, will be aware of climate change.
  - If hermit has been living in same place for long enough, has probably experienced climate change!!!
- Climate change will affect the ecology of the fish pond.
  - Will a warmer pond lead to faster fish growth?
  - Will a warmer pond lead to more food for fish to eat?
  - Will a warmer pond increase risk of eutrophication and collapse of fish?
- System is complex. Many possible outcomes. Unlikely to be able to identify all possibilities.
- System is complex. Unlikely to be able to accurately guage probabilities of possible outcomes.
- This is uncertainty!!!

Slide 68: Uncertainty

- There are two buckets with 100 ping pong balls in them.
  1. Bucket #1 has 50 red balls and 50 black balls.
  2. Bucket #2 has a mixture of red and black balls, numbers of each unknown.
- If you will be given $100 if you pull a red ball out of the bucket, which bucket will you choose?
  1. Bucket #1
  2. Bucket #2

Slide 69: Uncertainty

- There are two buckets with 100 ping pong balls in them.
  1. Bucket #1 has 50 red balls and 50 black balls.
  2. Bucket #2 has a mixture of red and black balls, numbers of each unknown.
- If you will be given $100 if you pull a black ball out of the bucket, which bucket will you choose?
  1. Bucket #1
  2. Bucket #2

Slide 70: Ellsberg Paradox

- There are two buckets with 100 ping pong balls in them.
  1. Bucket #1 has 50 red balls and 50 black balls.
  2. Bucket #2 has a mixture of red and black balls, numbers of each unknown.
- If you choose bucket #1 when a red ball earns $100, this implies that you believe odds of picking a red ball from bucket #2 is less than even.
- If you think the odds of choosing a red ball from bucket #2 are less than even, then you must think the odds of drawing a black ball are better than even.
- Majority of people choose bucket #1, whether the prize comes from a red or black ball.
- People generally prefer known odds to unknown odds.
- People are generally ***uncertainty averse***.

Slide 71: Handing Uncertainty

- With risk, can maximize some function based on the distribution of outcomes.
  - Maximize the expected value, when don't care about the variability.
  - Maximize the expected value less a penalty for volatility. A way of accounting for risk aversion.
- With uncertainty don't know the distribution.
- People may respond to their perception of best and worst case outcomes that follow from different options.
  - People who follow a ***maximin*** (maximize the minimum) approach choose options that have the least bad (maximum) worst possible outcome (the minimum). Kind of like risk averse.
  - People who follow a ***maximax*** (maximize the maximum) approach choose options that best possible (maximum) best outcome (the maximum). Kind of like risk loving.
  - Other strategies people follow fall between these approaches, involving some balancing of the extremes and other anticipated outcomes.

Slide 72: Handing Uncertainty

- Something about risk and uncertainty faced by entire groups.
- Something about precautionary principle.

content...

Chapter 5

Social Ecological Systems #2

Sustainability challenges exist because of the interaction between humans and the environment they inhabit.

Lets consider a system that incorporates humans and the environment.

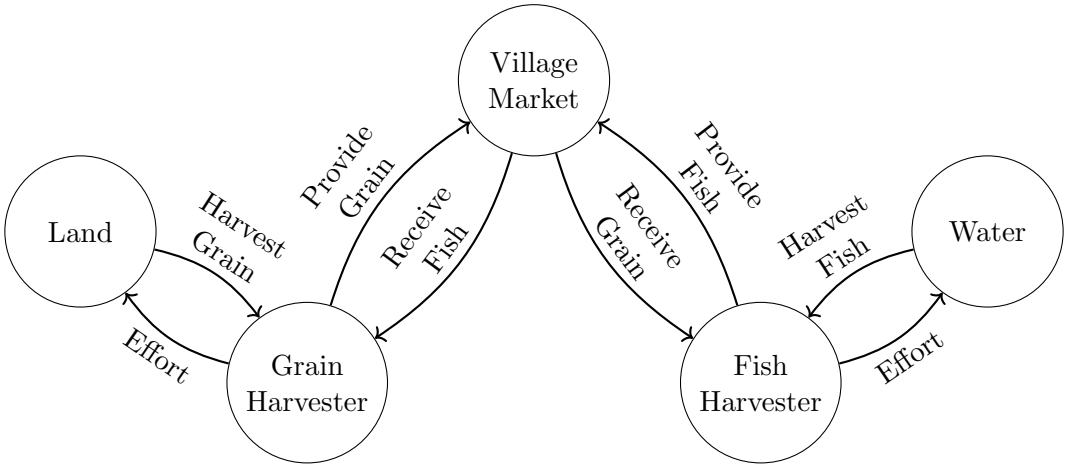
A simple example is a 'stylized' hunter gatherer model. In this case, our hunters will harvest fish, building on the model considered earlier.

Slide 73: The Hunter Gatherer Village

- The Question: ?
  - 1

An apparently simple mode can become complex pretty quickly.

Slide 74: The Hunter Gatherer Village



Slide 75: The Hunter Gatherer Village

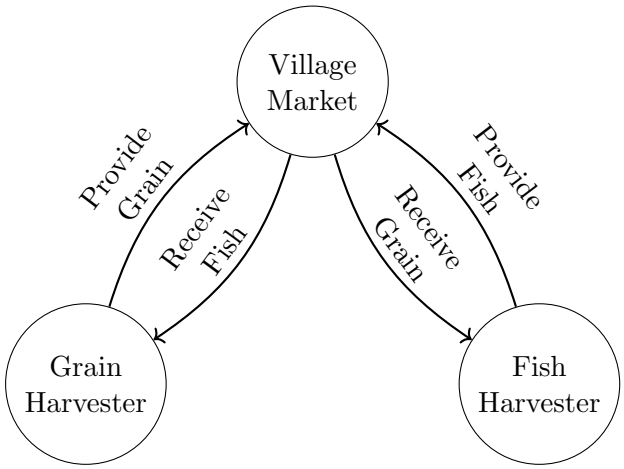
- The grain subsystem
  - Grain grows on the land
  - Grain harvesters exert effort that translates into harvested grain.
- The fish subsystem
  - Fish grow in the water
  - Fish harvesters exert effort that translates into harvested fish.
- The Village system
  - Fish harvesters bring fish to the village, receive grain.
  - Grain harvesters bring grain to the village, receive fish.
  - How is the exchange of fish for grain managed?

For the fish harvester, the travel distance before fish can be harvested is the same for every trip. For the grain harvester, the village is located among land that can produce grain. The harvester does not need to travel far if the land near the village has grain to harvest. However, if the land near the village has been depleted, then the grain harvester has to travel out from the village before any grain can be harvested, and has to travel further on the return trip.

How would things change if the amount of effort needed to travel to land that has grain? How would things change if the effort required to harvest a unit of grain is reduced? These are the impacts of technological change.

Slide 76: The Village Market

- How do fish and grain harvesters exchange their fish and grain?
- A collective? From each according to their means, to each according to their needs?
- A place where individuals trade for their own gain?



Figuring out how to represent the exchange of fish for grain is a tricky problem. What are the motivations that drive the decisions of these two people? Before delving into this problem, we can consider how their efforts may be combined to yield different combinations of fish and grain. For this exercise, imagine that there is enough fish and grain to share between the villagers so that they can live at least long enough to survive the trip to and from their respective harvest areas. If this is all they have, then they can travel out and back, without doing any harvesting or gathering, and upon return they would die, marked at E_1. Not ideal.

If there is sufficient food to allow them to survive beyond the travel time, then we can examine the combinations of fish and grain that can be produced by dividing the effort in different ways between fishing and gathering. In the figure, each line represents the combinations that can be produced for a given effort level. These curves are known as the ***production possibilities frontier (PPF)***. It is a frontier because it is what could be produced if all effort is used.

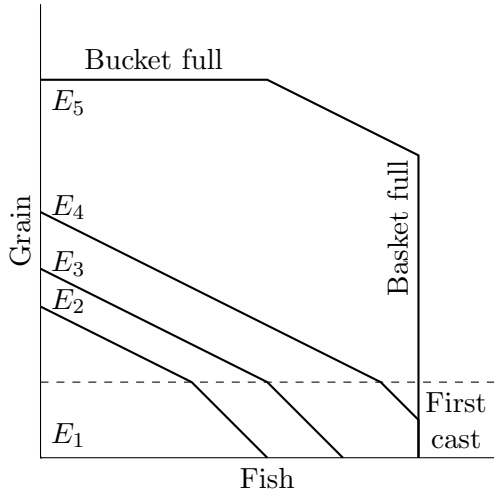
For the curves represented by E_2 and E_3, if the effort is reallocated from harvesting fish to gathering grain, the amount of fish harvested decreases and the amount of grain harvested increases.

With the PPF labeled E_4, we hit the limit of what the fish basket can hold. In effect, we 'throw away' part of the total effort, as we are dedicating it to catching fish, which we cannot take back to the village. Since our first cast is the most productive, we are in fact throwing away our most productive effort when running up against this constraint. If this seems strange, think of the order of events. The fish harvester cannot devote effort to later catches until they have made our first cast.

The PPF labeled E_5 corresponds to an effort level where one of the grain bucket or fish basket is full for most of the effort divisions. There is in effect a lot of wasted effort here, as it is being applied to activities which cannot result in any more grain or fish being brought back to the village. We might think of this wasted effort as leisure. If there is enough food in the village to sustain these harvesters and gatherers for this many days, maybe it would be better for them to stay in the village and enjoy themselves while eating the stockpiled food, heading out when they have enough food left to fill their basket or bucket and bring it back to the village.

Slide 77: Production Possibilities

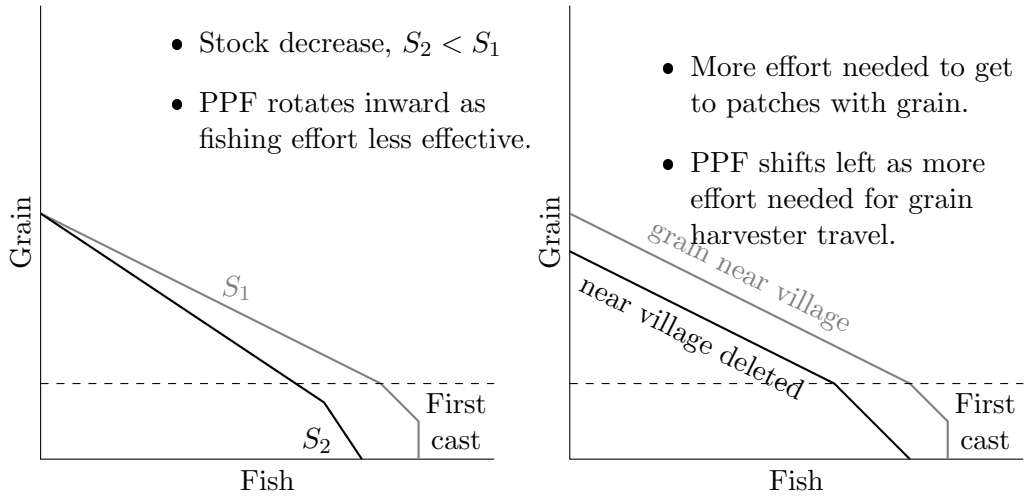
- Split total effort between fish and grain.
- $E_5 > E_4 > E_3 > E_2 > E_1$
- At E_1, no fish caught or grain gathered.
- At E_5, most effort not used, as bucket or basket full.



When the production functions were described above, some aspects relevant to sustainability were illustrate that do not show up here. For fishing, if harvesting reduces the fish stock, then the production function slope becomes flatter. It will take more effort, more days spent fishing, to fill up the fish basket. This shows up in the PPF as a rotation inward around the intercept with the Grain axis. The decline in the fish stock does not affect the amount of grain that can be gathered, if all the effort is devoted to gathering grain. However, the amount of grain production that is given up to harvest another fish increases.

When the grain available in patches near the village is depleted, more effort must be spent on travel. This reduces the total effort available for both activities, if we start from the same premise where we only examine how to reallocate effort after enough food is devoted to enabling both the fish harvester and the grain gatherer to travel to and from their respective activity locations. The PPF shifts left, lowering the intersection with the Grain axis. Since the total effort to divide is reduced, there is no longer enough effort to fill the fish basket, even when all the effort is put toward fishing.

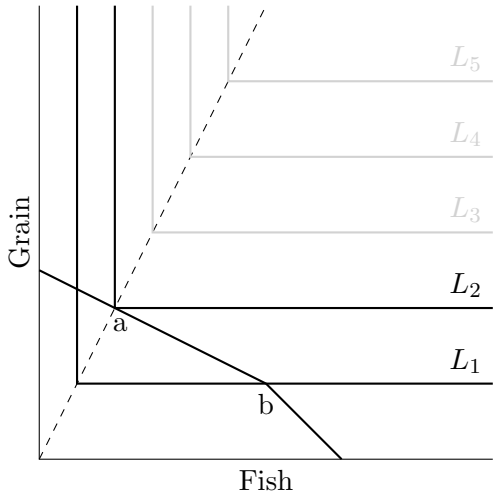
Slide 78: Production Possibilities



The production possibilities frontier tells us what combinations of fish and grain can be produced by the villagers, for different levels of effort. The biological reality of our villagers requires that the consume a particular bundle of fish and grain each day. If we put the biology together with the production possibilities frontier, we can see how to allocate the effort to provide the most days of life for the villagers.

Slide 79: Production Possibilities

- Lifespans L_1 and L_2 are attainable.
  - At **b** attain L_1.
  - At **a** attain L_2.
  - Same effort level.
- Lifespans L_3, L_4 and L_5 are not attainable.



Putting together the human needs and the production possibilities lets us identify something we might see as optimal. It seems reasonable to think that dividing up the resources that the villagers have in such a way as to maximize the number of days that they continue living would be a reasonable objective.

Can we expect this outcome from people who interact independently?

Slide 80: The Village Market

- 1.

